

One Judge After Another: Mechanistic Fairness of Protected-Attribute Bias in Language Reward Models

Cedric Kopp Frankfurt AI Safety — Reward-Model-Bias Working Group

Draft — July 10, 2026

Abstract

Language reward models (RMs) are the load-bearing evaluator of modern alignment: one preference model is reused across RLHF, best-of- N selection, and data filtering, so a bias in that single component propagates and compounds. We ask whether RMs inherit EU-protected demographic and intersectional biases; whether such bias is *low-complexity* (a near-linear direction that null-space projection removes at near-zero cost to accuracy) or *high-complexity* (entangled with legitimate signal); and, crucially, whether removal is *robust*. Extending the mechanistic reward-shaping method of Fein et al. [3] to protected attributes, and adding a closed-form concept-erasure test (LEACE [1] followed by a non-linear probe, the TACo signature of entanglement [8]), we run matched-pair contrastive experiments across three EU AI Act high-risk domains: credit, hiring, and essay grading. Preliminary results on a 0.6B RM show that demographic biases are large and behaviourally removable from the reward scalar, yet the underlying concept remains *non-linearly recoverable after optimal linear erasure*: null-space projection fixes the reward, not the representation. On essay grading the reliability-harm metric becomes interpretable, and a novel *standpoint-credibility* effect emerges, in which the RM rewards an identical argument more when it is voiced from a marginalized standpoint. We are scaling the study to eleven production reward models from 0.6B to 70B.

1 Introduction

The reward model is the most reused and least directly observed component of the alignment stack. A single preference model, trained once from human comparisons, is subsequently consulted as the arbiter of quality across RLHF policy optimization, best-of- N inference-time selection, iterative rejection-sampling data filtering, and automated evaluation. A systematic error in that one component is therefore not a local defect but an upstream point of failure whose effects surface, and compound, across many downstream behaviors. The EU AI Act’s high-risk regime (Annex III) names recruitment and creditworthiness as high-risk uses and requires technical robustness against biased outcomes on protected characteristics [2], yet most fairness auditing targets the deployed policy’s outputs rather than the evaluator that shaped them. Fein et al. [3] supply a mechanistic instrument that, for capability- and style-related biases, distinguishes biases removable by a cheap post-hoc edit from those entangled with legitimate signal, but they never apply it to protected attributes. This paper carries that low- versus high-complexity lens into the protected-attribute setting and strengthens it with a concept-erasure robustness check.

Research questions. **RQ0** (replication): do the base-paper interventions reproduce, and does the stronger LEACE/non-linear-probe test alter the low/high-complexity verdict? **RQ1**: which EU-protected attributes does an reward model exhibit a measurable bias for on otherwise-identical high-risk content, and is the intersectional effect distinct and non-additive? **RQ2**: is that bias low- or high-complexity, distinguishing the reward-scalar level from the representation level? **RQ3**: does mechanistic reward shaping reduce the harm without degrading legitimate accuracy? **RQ4**: is the removal robust, or does the concept leak back — the reward-vs-representation gap?

2 Related Work

The base paper. Fein et al. [3] measure biases in high-quality RMs and classify failures by complexity: low-complexity biases are approximately linear directions removed by a difference-of-means probe plus null-space projection, whereas high-complexity biases (sycophancy, model-style) are co-linear with useful signal and resist single-direction removal.

Demographic bias in RMs. Kumar et al. [5] introduce prefix bias and the auto- and cross-influence metrics; Song et al. [7] study group-level demographic parity across RMs. Neither asks whether the bias is linearly removable.

Concept erasure. LEACE [1] is a closed-form, provably optimal linear erasure; the TACo result [8] shows a concept can survive optimal linear erasure yet remain recoverable by a non-linear probe, which we adopt as the operational signature of entanglement; the lineage traces to iterative null-space projection [6].

Audit design. Our matched-pair construction follows the correspondence-audit lineage [9, 4].

Gap. No prior work applies the mechanistic linear / null-space lens to protected-attribute bias in the reward model itself.

3 Method

Design grammar. Every effect is measured on matched pairs that hold content fixed and vary only the target attribute, so content-driven activation cancels in the difference. We report two metrics [5]: auto-influence (preference shift on equal-quality items) and cross-influence (accuracy degradation when the attribute is attached to the weaker option, a reliability harm that maps onto the Act’s concern with biased false positives and negatives). Each attribute is tested in an *explicit* (stated) and a *proxy* (inferred) encoding.

Mechanistic step. From the matched pairs we estimate a difference-of-means direction and sweep the null-space projection strength α ; a bias is *reward-scalar* low-complexity if nulling removes it at near-zero cost to legitimate accuracy.

Representation-level test. Because null-space projection and a linear probe only interrogate the linear subspace, we additionally apply LEACE (optimal linear erasure) and then train a non-linear (MLP) probe on the erased representation. A concept that is linearly erasable but non-linearly recoverable is entangled (high-complexity) regardless of what the reward scalar shows — the discriminator at the heart of this work.

4 Data-Creation Pipeline (Overview)

A single matched-pair generator serves all domains through a domain registry. Credit decodes German Credit [10] into neutral financial profiles; hiring uses synthetic CVs carrying a strong/weak qualification label; education grades *real* student essays (PERSUADE 2.0 and ASAP-AES), holding the essay body byte-identical and injecting a single demographic marker as a header. Markers are single-axis (sex, age, family status, ethnicity via a name proxy, and education stage) in explicit and proxy encodings, with an intersectional anchor that composes the marginal clauses so the intersection direction can be tested against the sum of its marginals. A Tier-1 structural gate enforces an exact single-slot difference and length and readability parity, discarding failures. Full detail in Appendix A.

5 Experimental Setup

Models. We evaluate a ladder of eleven production reward models spanning 0.6B to 70B and the Llama, Qwen, Gemma, and DeBERTa families, chosen to disentangle *dataset-driven* from *architecture-driven* bias: it includes a same-recipe 8B/70B pair (for a controlled size comparison) and same-data / different-architecture pairs. Results reported here are on Skywork/Skywork-Reward-V2-Qwen3-0.6B; the larger models are in progress.

Metrics. auto-influence, cross-influence, an additivity (cosine) test for the intersection, a decision-response discriminatory-win-rate, a reasoning correctness $\times$ conclusion 2×2 , and linear versus non-linear probe recoverability after LEACE. Each cell uses $n=200$ held-out matched pairs, with robustness across encoding, template, and injection position, under a scalar matched-pair readout (primary) and a forced-choice pairwise cross-check.

6 Results

All values are on Skywork/Skywork-Reward-V2-Qwen3-0.6B; the larger reward models are in progress. Numbers pulled from the experiment JSONs are typeset from single-source macros and never hand-edited.

6.1 Direct scoring (auto-influence)

On the hiring (CV) domain, explicit-sex auto-influence is 1.00, collapsing to 0.11 when the difference-of-means direction is nulled; the sex $\times$ age $\times$ family intersection behaves likewise ($1.00 \rightarrow 0.01$) and is only *partially additive* (cosine with the summed marginals 0.66), i.e. a genuine interaction. Credit replicates the pattern (sex $1.00 \rightarrow 0.06$). A robustness battery reveals an encoding twist on CVs: the RM reacts more to realistic *proxy* cues than to bald statements for age and family status (age 0.23 explicit vs 0.99 proxy; family 0.04 vs 0.91). All effects are behaviourally low-complexity: the α -sweep drives them to near zero.

6.2 Cross-influence (reliability harm)

On credit and CV the metric is not yet interpretable on Skywork/Skywork-Reward-V2-Qwen3-0.6B (acc_baseline 0.59 on CV): a general chat RM does not track loan risk or CV quality, so the strong/weak label is not a valid ground truth. On *essay grading*, where the RM does track answer quality, cross-influence becomes interpretable (acc_baseline 0.72 on ASAP, 0.88 on PERSUADE), though the observed shift is a symmetric marker main-effect rather than a protected-attribute-specific harm at this scale.

6.3 Which biases, across domains

The education domain replicates the direct-scoring picture on real essays: stating sex is a large bias (auto-influence 0.88 on ASAP, 0.95 on PERSUADE), and an education-stage proxy is the largest single effect (0.96 / 0.97 — the RM rewards the identical essay far more when it is attributed to a university student rather than a school pupil), while name-based ethnicity is weak and template-unstable.

6.4 Reasoning vs. fairness

Modelling the RM’s scored *response* rather than the applicant, the RM rewards factually correct causal reasoning (+1.45 on a parental-leave premise) essentially equally on a non-demographic commute control (+1.46): a domain-general reasoning-quality effect, not demographic fairness. Consistently, the decision-response discriminatory-win-rate is near zero for sex (0.00) and age (0.00) but rises on the legally salient family and intersectional axes (0.39, 0.32).

6.5 Standpoint-credibility (positioned arguments)

When the same argument is grounded in a stated first-person identity, the RM rewards it more from a marginalized standpoint: the intersectional identity gap (reward of the marked vs. reference pole, relative to a no-standpoint baseline) is about +0.70, with sex +0.20, race +0.22, class +0.36, and origin +0.28, while genuinely neutral non-demographic controls (hobby, pet, region) sit at approximately zero. The effect is robust across injection position, corpus, three paraphrasings, and a neutral (non-endorsing) stance, and sits on top of a general reward for any lived-experience appeal.

6.6 Complexity: LEACE + non-linear probe

The decisive test: after LEACE, the linear probe for the reasoning-correctness concept drops to 52% (chance, by construction) while an MLP still recovers it at 100%. The concept is linearly erasable but non-linearly recoverable, the TACO signature of entanglement. The reward-scalar low-complexity reading is therefore incomplete: null-space projection collapses the bias’s contribution to the reward, but the concept remains present in the representation.

7 Discussion

The reward-vs-representation gap. Our central mechanistic finding is that a bias can be simultaneously behaviourally removable and representationally entangled: null-space projection zeroes its contribution to the (linear) reward head, yet the concept survives optimal linear erasure and is recovered by a non-linear probe. Debiasing the reward is therefore not the same as debiasing the model. For compliance-grade mitigation this is a caution: a cheap post-hoc edit can make an evaluator appear unbiased on the metric while the discriminating information remains available to any downstream non-linear computation, so the *robustness* of the intervention, not its immediate effect, is the load-bearing question.

Limitations. Results so far are on a single 0.6B RM and use templated matched-pair items; the scaling study across eleven RMs up to 70B, and the standpoint arm on a topic-natural corpus, are in progress.

8 Conclusion

Protected-attribute bias in language reward models is, on our preliminary evidence, behaviourally removable from the reward scalar yet representationally entangled: nulling fixes the reward, not the representation. Alongside this we obtain an interpretable reliability-harm metric on essay grading and a novel pro-marginalized standpoint-credibility effect. Whether these findings, and the robustness of the mechanistic intervention, hold on the frontier-scale reward models actually deployed is the open question our ongoing scaling study addresses.

References

- [1] N. Belrose, D. Schneider-Joseph, S. Ravfogel, R. Cotterell, E. Raff, and S. Biderman. LEACE: Perfect linear concept erasure in closed form. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. arXiv:2306.03819.
- [2] European Parliament and Council. Regulation (eu) 2024/1689 (artificial intelligence act). Official Journal of the European Union, 2024.
- [3] D. Fein, M. Lamparth, V. Xiang, M. J. Kochenderfer, and N. Haber. One bias after another: Mechanistic reward shaping and persistent biases in language reward models. In *International Conference on Machine Learning (ICML)*, 2026. arXiv:2603.03291.
- [4] A. Haim, A. Salinas, and J. Nyarko. What’s in a name? auditing large language models for race and gender bias, 2024. arXiv:2402.14875.

- [5] A. Kumar, Y. He, A. H. Markosyan, B. Chern, and I. Arrieta-Ibarra. Detecting prefix bias in llm-based reward models. In *ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, 2025. arXiv:2505.13487.
- [6] S. Ravfogel, Y. Elazar, H. Gonen, M. Twiton, and Y. Goldberg. Null it out: Guarding protected attributes by iterative nullspace projection. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2020. arXiv:2004.07667.
- [7] K. Song, J. Yao, R. Jiang, R. Chandra, and S. Zhang. Towards large language models that benefit for all: Benchmarking group fairness in reward models. In *Reinforcement Learning Conference (RLC)*, 2025.
- [8] TaCo Authors. Taco: Targeted concept erasure prevents non-linear classifiers from detecting protected attributes, 2024. arXiv:2312.06499.
- [9] A. Tamkin, A. Askill, L. Lovitt, et al. Evaluating and mitigating discrimination in language model decisions, 2023. arXiv:2312.03689; discrim-eval, CC-BY-4.0.
- [10] UCI Machine Learning Repository. Statlog (german credit data), 1994. CC-BY-4.0.

A Data-Creation Pipeline — Detailed Specification

A.1 Substrates

[**TODO**: Credit: German Credit [10] (CC-BY-4.0), decoded A-codes $\rightarrow$ neutral financial profile; sex+marital entanglement motivates synthetic injection. Hiring: synthetic CVs (CandidateRecord with a strong/weak **qualified** label, the quality ground truth for cross-influence). Education (grading, *built*): **real essay corpora** held byte-identical — PERSUADE 2.0 (CC-BY-NC-SA; argumentative, holistic 1–6 score) and ASAP-AES (Kaggle 2012 Hewlett) — loaded as **EssayRecords**; the demographic marker is injected only as a header clause (approach A1). **high_quality** is thresholded from the holistic score (strong = top tier / weak = bottom, middle dropped) — the cross-influence quality ground truth, which the RM *does* track for essays (unlike credit/CV).]

A.2 Marker injection (single-axis)

[**TODO**: Neutral slot-templated renderer with one **{marker}** slot; sex/age/family-status + the $\text{sex} \times \text{age} \times \text{family}$ intersection; explicit and proxy (name / graduation-year / career-gap) encodings; the intersection clause composes the three marginal clauses (enables the additivity test). For the education domain (A1, header injected over real essays): **name** $\rightarrow$ sex + **ethnicity** (a name grid coded to origin, Haim et al. lineage [4]); and **grade_level** (7th-grade pupil vs final-year university student) as an age/education-stage proxy; plus explicit forms. Shared single-axis markers (also usable by credit/CV); the subject noun is parameterized (“the applicant” $\rightarrow$ “the student”). Disability deferred. **Positioned-argument (A2)** variant (**positionality.py**): instead of a header, inject one first-person *standpoint* sentence whose claimed identity is the swapped unit (“as {a woman / ... / a working-class Black woman} who has lived these realities firsthand, I am convinced. ...”), grounding the argument’s authority; the whole sentence is the strippable unit, and the insertion *position* (conclusion/opening/middle/random) is a parameter (index chosen once, reused for both poles). Non-demographic control axes (hobby/pet/region) and a no-positionality neutral baseline enable the main-effect vs identity-gap decomposition.]

A.3 Tier-1 structural validation gate

[**TODO**: Per matched pair: (1) exact single-slot difference (strip each marker clause $\rightarrow$ byte-identical remainders); (2) length parity ($|\Delta_{\text{char}}|, |\Delta_{\text{token}}|$ bounded); (3) readability parity ($|\Delta_{\text{Flesch}}|$ bounded). Over-generate, discard failures, record discard rate in the manifest.]

A.4 Manifest, loader, reproducibility

[**TODO:** Self-describing JSONL manifest (axis/encoding/template/intersectional cell/provenance); per-RM chat-template formatting applied at scoring time; datasets regenerable from seed. Domain registry (credit/cv/education) as the single dispatch point — adding a domain is one `DomainSpec` entry.]

A.5 Verdict + reasoning items (model-response arm)

[**TODO:** Authored fair/discriminatory/neutral/evasive verdicts (length-matched); the correctness×conclusion reasoning 2×2 with paraphrase pools + claim types + a non-demographic commute control.]